

PRÉPARATION DES DONNÉES

MATRICE ORIGINELLE

	gene A	gene B	gene N
cell 1	0	1	2
cell 2	7	30000	0
cell n	0	20	10000

FILTRE
CELLULES/GÈNES

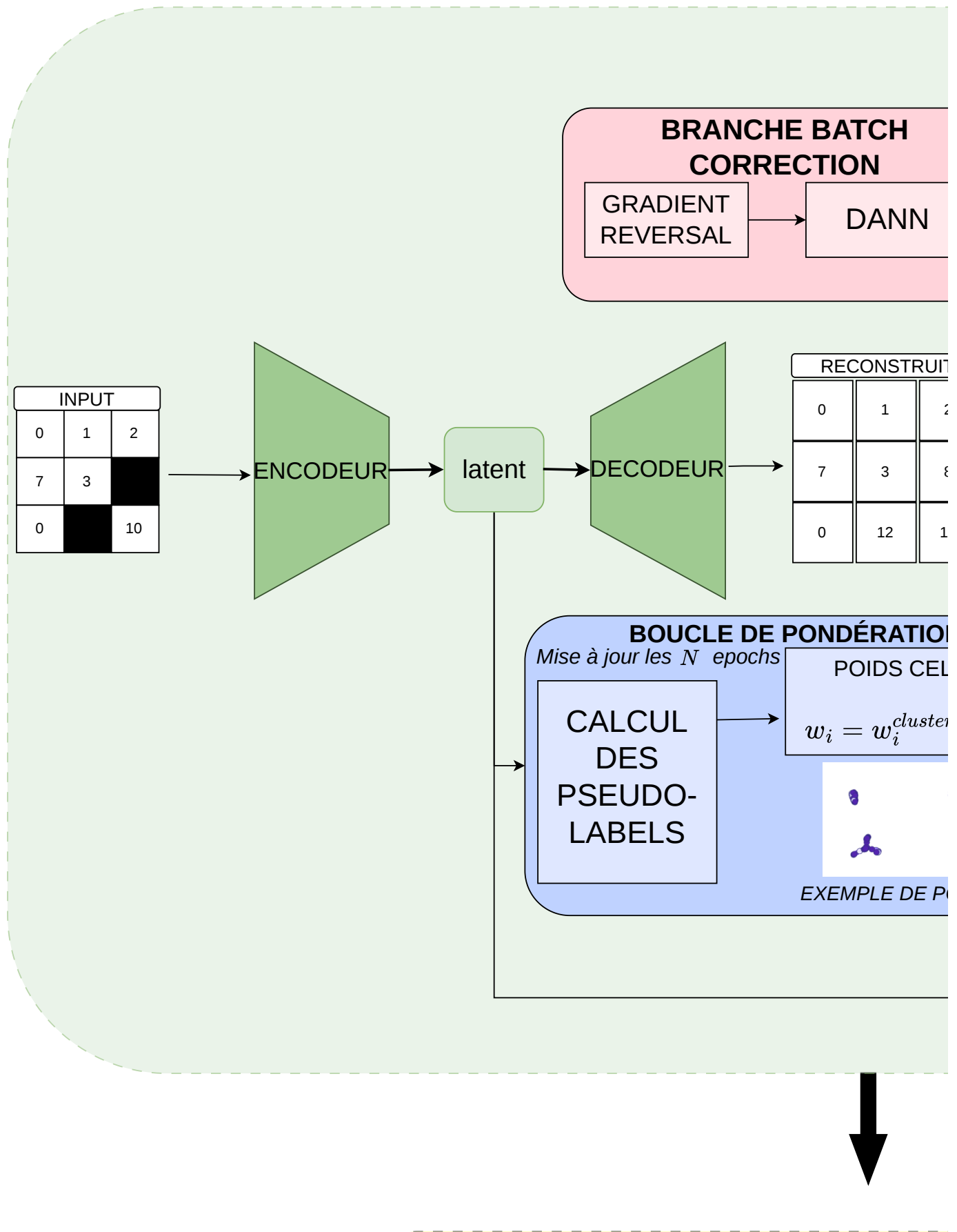
NORMALISATION

LOG1P

SÉLECTION HVG

SCALING / CLIPPING





ÉCHAUFFEMENT
RECONSTRUCTION
UNIFORME

PHASE PONDÉRÉE
PONDÉRATION DYNAMIQUE

MLP AUTOENCODEUR SYMÉTRIQUE

LOSS
BATCH

LOSS DE RECONSTRUCTION

Warm-up : MSE Uniform

Phase
pondérée

$$L_{recon,weighted} = \frac{1}{N} \sum_{i=1}^N w_i L_{recon,i}$$

+

TRIPLET
LOSS

T
2
8
.0

N
LULAIRES

$$r + w_i^{density}$$



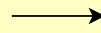
ONDÉRATION

SOTA CLUSTERING

ASSIGNATION DES CLUSTERS

SI LABEL
RÉASSIGNÉ

latent
final



cell_1	clust_0
cell_2	clust_1
cell_3	clust_1
cell_n	clust_2

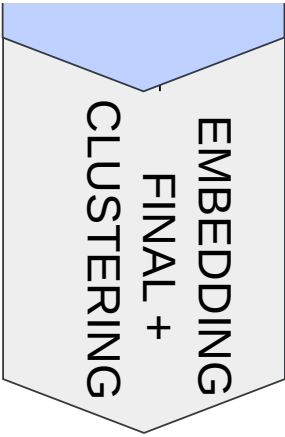
cell_1	
cell_2	
cell_3	
cell_n	



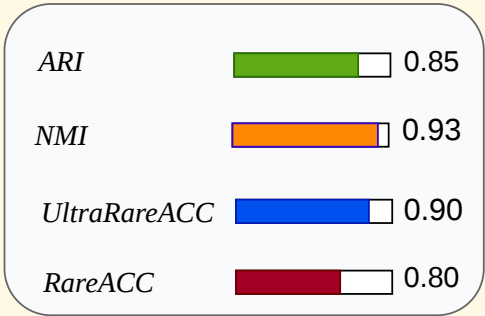
UMAPS

ELS DISPONIBLES :
GNATION DES LABELS

	clust_0	alpha
2	clust_1	beta
3	clust_1	beta
4	clust_2	gamma



ANALYSE



SCIB-E 0.75 ACC 0.97

MÉTRIQUES

